

NoSQL Unit 5

UNIT – V Topicwise Question Bank

1. Graph Databases: Introduction & Features

1 Mark Questions

Q1. Which is a key feature of graph databases? [1 Mark]

- a) fixed schema
- b) efficient storage and querying of relationships
- c) store data only in tabular format
- d) use of SQL for all queries

Q2. Which is the best definition for the graph database? [1 Mark]

- A) A system that organizes data in collections and documents.
- B) A system that stores information in tabular formats.
- C) A database built on the concept of graph theory using nodes and edges.
- D) A relational database system optimized for joins.

Q3. What makes Neo4j distinct as a database system? [1 Mark]

- A) It uses JSON-based querying.
- B) It integrates natively with MySQL.
- C) It leverages graph structures and the Cypher query language.
- D) It's primarily used for time-series analysis.

6 Mark Questions

Q4. Compare Relational Databases, Graph Databases, Key-Value Stores, and Column-Family Stores focusing on their schema design, query languages, use cases and performance. [6 Marks]

Q5. Explain the fundamental idea behind graph databases and highlight the key differences between graph databases and relational database systems. [6 Marks]

2. Graph Data in Relational Database / Querying Graph Data using MySQL

6 Mark Questions

Q6. Compare Relational Databases, Graph Databases, Key-Value Stores, and Column-Family Stores focusing on their schema design, query languages, use cases and performance. [6 Marks]

Q7. Explain the fundamental idea behind graph databases and highlight the key differences between graph databases and relational database systems. [6 Marks]

3. Neo4j in NoSQL Space

1 Mark Questions

Q8. Which of the following is NOT a common data model used in NoSQL databases? [1 Mark]

- a) Key-value store
- b) Column-family store
- c) Document store
- d) Relational table store

Q9. What makes Neo4j distinct as a database system? [1 Mark]

- A) It uses JSON-based querying.
 - B) It integrates natively with MySQL.
 - C) It leverages graph structures and the Cypher query language.
 - D) It's primarily used for time-series analysis.
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4. Key-Value Stores

1 Mark Questions

Q10. Which REDIS datatype is best suited to store a collection of unique values? [1 Mark]

- a) string
- b) list
- c) set
- d) hash

6 Mark Questions

Q11. Compare Relational Databases, Graph Databases, Key-Value Stores, and Column-Family Stores focusing on their schema design, query languages, use cases and performance. [6 Marks]

5. Column-Family Stores

1 Mark Questions

Q12. Which of the following is NOT a common data model used in NoSQL databases? [1 Mark]

- a) Key-value store

- b) Column-family store
- c) Document store
- d) Relational table store

6 Mark Questions

Q13. Compare Relational Databases, Graph Databases, Key-Value Stores, and Column-Family Stores focusing on their schema design, query languages, use cases and performance. [6 Marks]

6. Data Modeling in Neo4j

1 Mark Questions

Q14. Why would a Neo4j node have both :Customer and :VIP labels? [1 Mark]

- A) To increase indexing performance by duplicating labels
- B) To allow querying the node as either :Customer or :VIP
- C) Because each node must have a minimum of two labels
- D) To prevent the node from being matched in generic queries

4 Mark Questions

Q15. Write Cypher queries to create 5 Person nodes with properties (user_ID, name, age, email, location), connect them with FRIEND relationships, add a city property to each, remove the city property from persons living in "Bangalore", and delete persons older than 25. [4 Marks]

Q16. Write Cypher queries to create 5 Actor nodes and 5 Movie nodes, connect them with ACTED_IN relationships while setting a role property for each relationship. Find movies whose titles start

with "F", count the number of movies each actor has acted in, and return the top actors ranked by their number of movies using ORDER BY, LIMIT, and SKIP. [4 Marks]

Q17. Write Cypher queries to create City nodes for the list ["Berlin", "Delhi", "Madrid", "Rome", "Osaka"] using FOREACH, assign a region property based on city name (e.g., "Asia" for "Delhi" and "Osaka"). Find and return cities starting with "D", delete cities in the "Europe" region, and finally count and return the number of remaining cities as city_count. [4 Marks]

Q18. Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order. [4 Marks]

7. Neo4j CQL Read/Write Clauses

A. MATCH Clause

1 Mark Questions

Q19. In Cypher (Neo4j), which clause is used to find patterns in the graph? [1 Mark]

- A) MATCH
 - B) FIND
 - C) QUERY
 - D) SEARCH
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B. SET Clause

6 Mark Questions

Q20. Write a Cypher query using Neo4j to add nodes using the SET, SKIP, Aggregation clause. Explain the syntax and effect of the query. [6 Marks]

4 Mark Questions

Q21. Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order. [4 Marks]

C. DELETE Clause

1 Mark Questions

Q22. Which neo4j clause is used to remove properties or labels from nodes? [1 Mark]

- a) delete
- b) remove
- c) foreach
- d) limit

4 Mark Questions

Q23. Write Cypher queries to create City nodes for the list ["Berlin", "Delhi", "Madrid", "Rome", "Osaka"] using FOREACH, assign a region property based on city name (e.g., "Asia" for "Delhi" and "Osaka"). Find and return cities starting with "D",

delete cities in the "Europe" region, and finally count and return the number of remaining cities as city_count. [4 Marks]

Q24. Write Cypher queries to create 5 Person nodes with properties (user_ID, name, age, email, location), connect them with FRIEND relationships, add a city property to each, remove the city property from persons living in "Bangalore", and delete persons older than 25. [4 Marks]

D. REMOVE Clause

1 Mark Questions

Q25. Which neo4j clause is used to remove properties or labels from nodes? [1 Mark]

- a) delete
- b) remove
- c) foreach
- d) limit

4 Mark Questions

Q26. Write Cypher queries to create 5 Person nodes with properties (user_ID, name, age, email, location), connect them with FRIEND relationships, add a city property to each, remove the city property from persons living in "Bangalore", and delete persons older than 25. [4 Marks]

E. WHERE Clause

4 Mark Questions

Q27. Write Cypher queries to create City nodes for the list ["Berlin", "Delhi", "Madrid", "Rome", "Osaka"] using FOREACH, assign a region property based on city name (e.g., "Asia" for "Delhi" and "Osaka"). Find and return cities starting with "D", delete cities in the "Europe" region, and finally count and return the number of remaining cities as city_count. [4 Marks]

Q28. Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order. [4 Marks]

Q29. Write Cypher queries to create 5 Actor nodes and 5 Movie nodes, connect them with ACTED_IN relationships while setting a role property for each relationship. Find movies whose titles start with "F", count the number of movies each actor has acted in, and return the top actors ranked by their number of movies using ORDER BY, LIMIT, and SKIP. [4 Marks]

F. Count Function

4 Mark Questions

Q30. Write Cypher queries to create 5 Actor nodes and 5 Movie nodes, connect them with ACTED_IN relationships while setting a role property for each relationship. Find movies whose titles start with "F", count the number of movies each actor has acted in, and return the top actors ranked by their number of movies using ORDER BY, LIMIT, and SKIP. [4 Marks]

Q31. Write Cypher queries to create City nodes for the list ["Berlin", "Delhi", "Madrid", "Rome", "Osaka"] using FOREACH,

assign a region property based on city name (e.g., "Asia" for "Delhi" and "Osaka"). Find and return cities starting with "D", delete cities in the "Europe" region, and finally count and return the number of remaining cities as city_count. [4 Marks]

Q32. Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order. [4 Marks]

G. ORDER BY Clause

1 Mark Questions

Q33. In HiveQL, what is the main difference between JOIN and GROUP BY? [1 Mark]

- a) JOIN combines rows from two or more tables; GROUP BY aggregates rows based on column values
- b) JOIN filters rows; GROUP BY sorts rows
- c) JOIN sorts rows; GROUP BY combines rows
- d) JOIN aggregates rows; GROUP BY filters rows

4 Mark Questions

Q34. Write Cypher queries to create 5 Actor nodes and 5 Movie nodes, connect them with ACTED_IN relationships while setting a role property for each relationship. Find movies whose titles start with "F", count the number of movies each actor has acted in, and return the top actors ranked by their number of movies using ORDER BY, LIMIT, and SKIP. [4 Marks]

Q35. Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order. [4 Marks]

H. LIMIT Clause

6 Mark Questions

Q36. Write a Cypher query using Neo4j to add nodes using the SET, SKIP, Aggregation clause. Explain the syntax and effect of the query. [6 Marks]

4 Mark Questions

Q37. Write Cypher queries to create 5 Actor nodes and 5 Movie nodes, connect them with ACTED_IN relationships while setting a role property for each relationship. Find movies whose titles start with "F", count the number of movies each actor has acted in, and return the top actors ranked by their number of movies using ORDER BY, LIMIT, and SKIP. [4 Marks]

I. SKIP Clause

6 Mark Questions

Q38. Write a Cypher query using Neo4j to add nodes using the SET, SKIP, Aggregation clause. Explain the syntax and effect of the query. [6 Marks]

4 Mark Questions

Q39. Write Cypher queries to create 5 Actor nodes and 5 Movie nodes, connect them with ACTED_IN relationships while setting a role property for each relationship. Find movies whose titles start with "F", count the number of movies each actor has acted in, and return the top actors ranked by their number of movies using ORDER BY, LIMIT, and SKIP. [4 Marks]

8. String Functions

4 Mark Questions

Q40. Write Cypher queries to create 5 Actor nodes and 5 Movie nodes, connect them with ACTED_IN relationships while setting a role property for each relationship. Find movies whose titles start with "F", count the number of movies each actor has acted in, and return the top actors ranked by their number of movies using ORDER BY, LIMIT, and SKIP. [4 Marks]

Q41. Write Cypher queries to create City nodes for the list ["Berlin", "Delhi", "Madrid", "Rome", "Osaka"] using FOREACH, assign a region property based on city name (e.g., "Asia" for "Delhi" and "Osaka"). Find and return cities starting with "D", delete cities in the "Europe" region, and finally count and return the number of remaining cities as city_count. [4 Marks]

Q42. Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order. [4 Marks]

9. Aggregate Functions

6 Mark Questions

Q43. Write a Cypher query using Neo4j to add nodes using the SET, SKIP, Aggregation clause. Explain the syntax and effect of the query. [6 Marks]

4 Mark Questions

Q44. Write Cypher queries to create 5 Actor nodes and 5 Movie nodes, connect them with ACTED_IN relationships while setting a role property for each relationship. Find movies whose titles start with "F", count the number of movies each actor has acted in, and return the top actors ranked by their number of movies using ORDER BY, LIMIT, and SKIP. [4 Marks]

Q45. Write Cypher queries to create City nodes for the list ["Berlin", "Delhi", "Madrid", "Rome", "Osaka"] using FOREACH, assign a region property based on city name (e.g., "Asia" for "Delhi" and "Osaka"). Find and return cities starting with "D", delete cities in the "Europe" region, and finally count and return the number of remaining cities as city_count. [4 Marks]

Q46. Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order. [4 Marks]